

Session 3

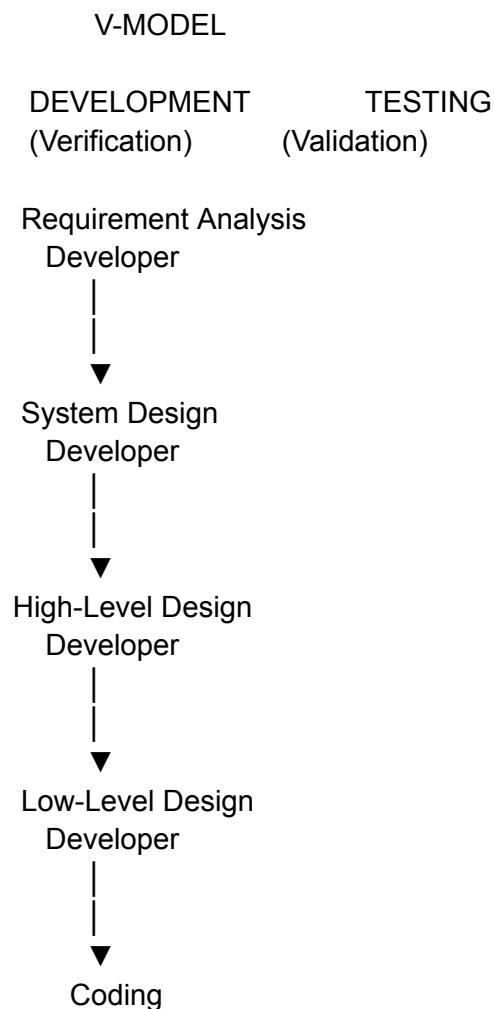
Software Testing Assignment

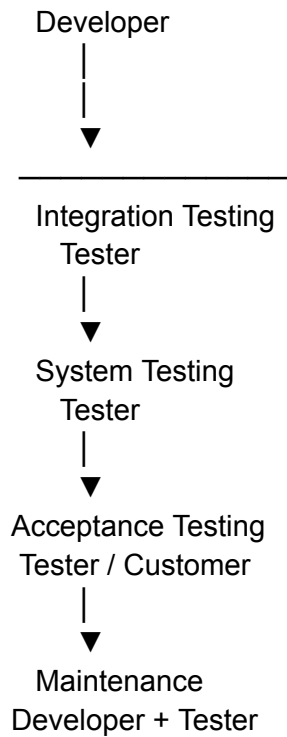
1. Draw the V-Model diagram on paper or using any digital tool, and clearly label each phase with both the developer and tester roles involved at each step.

1. V-Model Diagram with Developer and Tester Roles

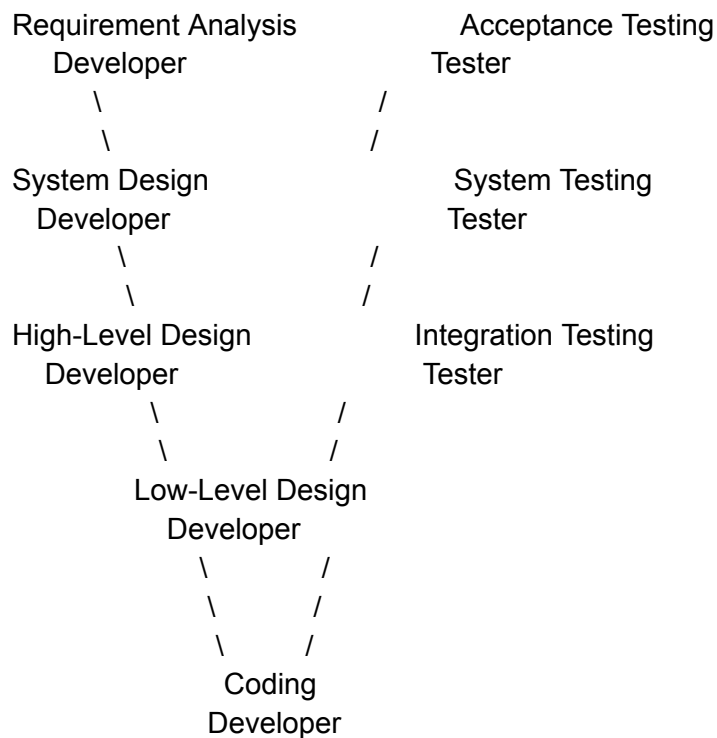
The **V-Model (Verification and Validation Model)** is a software development model where development activities are planned on the left side and corresponding testing activities are performed on the right side.

V-Model Diagram





Better V-Shape to draw on paper



Important: In the V-Model, testing is planned early. Each development phase has a corresponding testing phase.

2. Pick any app you use daily (like Zomato, Instagram, or Paytm) and write a short table mapping each SDLC phase (Requirement, Design, Coding, Testing, Deployment, Maintenance) to a real activity that would happen for that app.

SDLC Phases – Food Delivery App Example

Let's take **Zomato** as the example.

SDLC Phase	Real Activity in Zomato
1. Requirement	Collect requirements such as user login, restaurant search, food ordering, online payment, and order tracking.
2. Design	Design the app screens for login, restaurant list, menu, cart, payment, and order tracking.
3. Coding	Developers write code for login, restaurant search, cart, payment, and delivery tracking features.
4. Testing	Testers check login, food ordering, payment, coupons, notifications, and order tracking for defects.
5. Deployment	The completed application is released on the Play Store/App Store and made available to users.
6. Maintenance	Developers fix bugs, improve performance, add new features, and update the app based on user feedback.

**3. For each phase of the V-Model, list one specific testing activity that should be performed, using examples from a food delivery app (like Swiggy or Zomato).

Hint: For example, during 'System Testing', you might test if the order placement works end-to-end.**

Testing Activity for Each V-Model Phase

Using a **food delivery app like Zomato/Swiggy**:

V-Model Phase	Testing Activity	Example
Requirement Analysis	Requirement Validation	Check whether the requirement clearly states that users must be able to place food orders successfully.
System Design	System Test Planning	Prepare test cases for complete order flow, including restaurant selection, cart, payment, and delivery tracking.
High-Level Design	Integration Testing Planning	Plan how restaurant, cart, payment, and order-tracking modules will work together.
Low-Level Design	Unit Test Planning	Identify individual functions/modules that need testing, such as calculating the cart total.
Coding	Unit Testing	Test individual code functions, such as checking whether the total price is calculated correctly.
Integration Testing	Integration Testing	Check whether the cart correctly sends the order information to the payment system.
System Testing	System Testing	Test the complete process from selecting food to placing an order and receiving confirmation.
Acceptance Testing	User Acceptance Testing	Check whether the app satisfies real user requirements, such as successfully ordering food and tracking delivery.

Simple example:

User selects food → Adds to cart → Makes payment → Order confirmed → Delivery tracked

The tester checks whether this **complete flow works correctly**.

4. Compare the V-Model and RAD Model by listing 3 strengths and 2 weaknesses of each, focusing on how testing is handled in both models.

V-Model vs RAD Model

V-Model

3 Strengths

1. **Testing is planned early**
Testing activities are planned alongside development activities.
2. **Clear testing stages**
Each development phase has a corresponding testing phase, making testing systematic.
3. **Good documentation and quality control**
Requirements, designs, and test cases are clearly documented, making it easier to find defects.

2 Weaknesses

1. **Difficult to handle changing requirements**
If requirements change later, development and testing plans may need major changes.
 2. **Working software appears late**
Users generally don't get a working version until development is largely completed.
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RAD Model

RAD = Rapid Application Development

RAD focuses on **quick development, prototypes, user feedback, and repeated improvements.**

3 Strengths

1. **Testing happens repeatedly**
Each prototype or development cycle can be tested and improved quickly.
2. **Fast development**
Features can be developed and tested in short cycles.
3. **Easy to get user feedback**
Users can test prototypes and suggest changes before the final product is completed.

2 Weaknesses

1. **Testing can become rushed**
Because development is very fast, testers may have limited time for detailed testing.
 2. **Requires continuous user involvement**
Regular feedback and testing from users are important; without it, defects or wrong requirements may remain.
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Quick Comparison

Point	V-Model	RAD Model
Development style	Sequential	Iterative and fast
Testing	Planned early and performed systematically	Repeated throughout development
User feedback	Usually later	Frequent
Main advantage	Strong testing and documentation	Fast development and feedback
Main weakness	Difficult to change requirements	Testing may be rushed
Best suited for	Stable requirements	Projects where requirements may change